M-PSK

- Ortogonal modülasyon

Frekans kaydırmalı Anahtarlama (FKA)

Frequency Shift keying (FSK)

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_i t), \quad 0 \le t \le T, \quad i = 1, \dots, M = 2^k$$

$$f_i = f_c + (i-1)\Delta f$$

$$\Delta f = \begin{cases} \frac{1}{2T} & \text{Faz tahminli demod} \\ \frac{1}{T} & \text{// tahminsiz //} \end{cases}$$

$$\int_0^T s_i(t) s_j(t)\, dt = \begin{cases} 0 & i \neq j \\ E & i = j \end{cases}$$

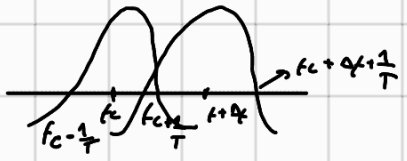
$\forall i \quad E_i = E$

$M \uparrow \rightarrow BG \uparrow \quad (BGV) \downarrow \quad \times P_b \downarrow$ Performans iyileşir $\Rightarrow$ güç verimli (Power efficient)

$$BGV = \frac{R_b \text{ bit/sn}}{BG \text{ Hz}}$$

Özel durum

$M = 2^1 \Rightarrow k = 1$ bit $\rightarrow$ ikili FRKA (İFRKA) Binary FSK (BFSK)



"1" $\rightarrow s_1(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t), \quad 0 \le t \le T$

"0" $\rightarrow s_2(t) = \sqrt{\frac{2E}{T}} \cos(2\pi (f_c + \Delta f)), \quad 0 \le t \le T$

$$\int_0^T s_1(t) s_2(t)\, dt = 0$$

$\downarrow$

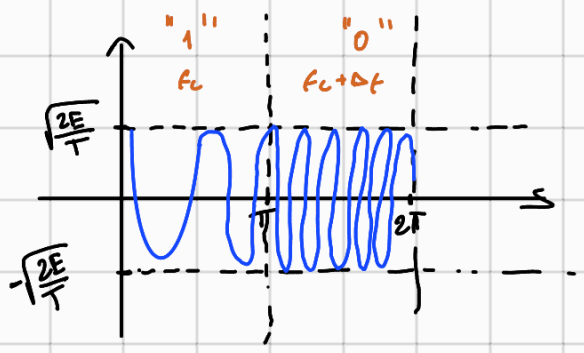
$\rho = 0$

$\downarrow$

$$P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right), \quad E_b = E$$

Unipolar signaling x carrier

bunaa benziyo



Performansı daha iyi

$\rightarrow BPSK = P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$

$BFSK: P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$

Bant genişliği daha fazla

M arttıkça FSK'in PSK'dan daha iyi bir performansa sahip olur

PSK'nin zarfı sabit değil, devreleri doymaya sokuyor.

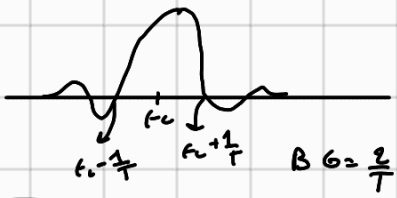
Sayfa 174 Şekil 4.5

Ortogonal modülasyon

Birbirine ortogonal 2 işaret

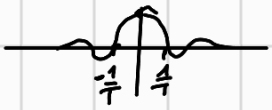
$$s_1(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t),\ 0 \le t \le T$$
$$s_2(t) = \sqrt{\frac{2E}{T}} \sin(2\pi f_c t),\ 0 \le t \le T \quad \Big\} f = 0$$

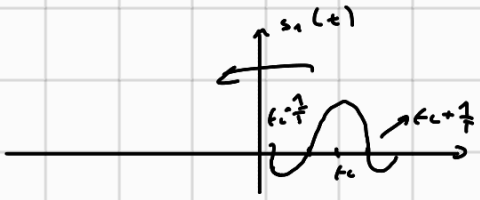
Bant genişliği: F_s kadar fazla



$BG = \frac{2}{T}$

$$s_1(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t),\ 0 \le t \le T$$
$$= \sqrt{\frac{2E}{T}} \Pi\left(\frac{t - T/2}{T}\right) \cos(2\pi f_c t)$$





Genlik kaydırmalı Anahtarlama

Amplitude Shift Keying (ASK) / PAM

$$s_i(t) = \sqrt{\frac{2}{T}} A_i \cos(2\pi f_c t),\ 0 \le t \le T,\ t = 1, \dots, M = 2^k$$

$BG = \frac{2}{T} nz$

$M \uparrow \Rightarrow$ BG aynı (BGV↑) $\times P_b \uparrow \Rightarrow$ BG verimli teknik

Örnek: $k = 2$ $M = 2^2 = 4$

		1. Teknik	2. Teknik
0	0	$\rightarrow A_1 = A$	$A_1 = A$
0	1	$\rightarrow A_2 = 2A$	$A_2 = -A$
1	0	$\rightarrow A_3 = 3A$	$A_3 = 2A$
1	1	$\rightarrow A_4 = 4A$	$A_4 = -2A$

$$E_{ort} = \frac{30A^2}{4} = 7,5A^2 \qquad E_{ort} = \frac{10A^2}{4} = 2,5A^2$$

$$E_i = \underbrace{\frac{2}{T} A_i^2}_{G=1} \frac{1}{2} \times \underbrace{T}_{\text{zaman}} = A_i \text{ Joule}$$

$E_i = A_i^2$

Simülasyon

Daha az enerjiyle aynı performans elde ederiz

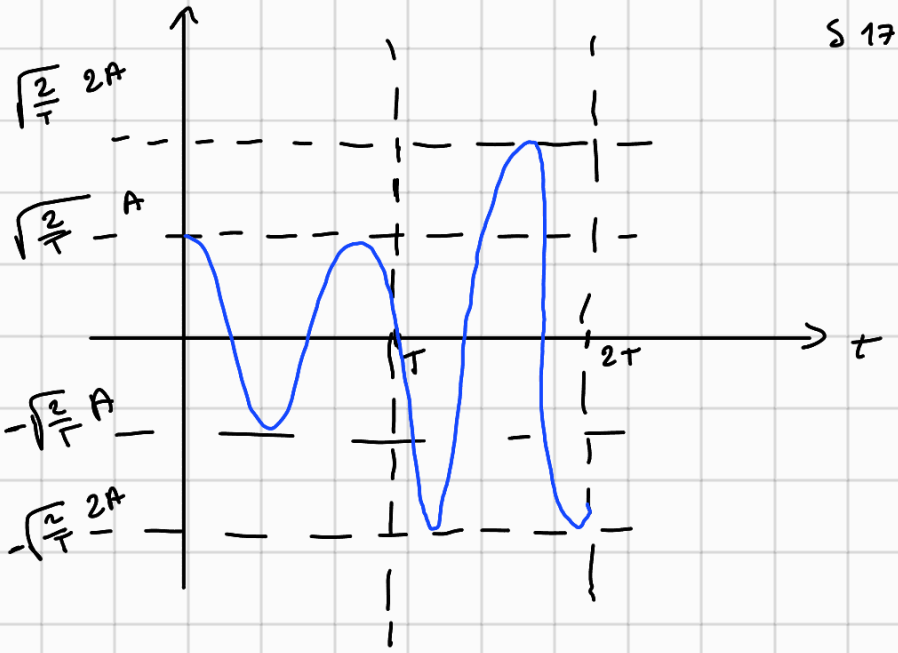
Özel durum: $M = 2 \rightarrow$ ikili GkA (iGkA)
Binary ASK (BASK)

$$s_1(t) = \sqrt{\frac{2}{T}} A \cos(2\pi f_c t),\ 0 \le t \le T$$
$$s_2(t) = -\sqrt{\frac{2}{T}} A \cos(2\pi f_c t),\ 0 \le t \le T$$
$A_2 = -A$

Binary ASK = Binary PSK

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right),\ E_b = A^2$$

$M = 2$ $\quad A_1 = A \;,\; A_2 = 2A$



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Özel durum

On off keying (OOK) / Var-yok anahtarlama (VYA)

$s_1(t) = \sqrt{\frac{2}{T}}\,A\cos(2\pi f_c t)\;,\; 0 \le t \le T$

$s_2(t) = 0\;,\; 0 \le t \le T$ $\quad \rightarrow \sqrt{\frac{2A^2}{T}} = \sqrt{\frac{2E}{T}}\;,\; E = A^2$

Unipolar signaling x carrier

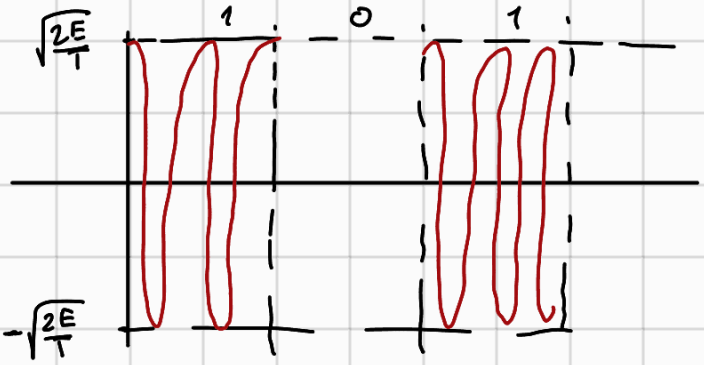
$\int_0^T s_1(t)\,s_2(t)\,dt = 0 \Rightarrow \rho = 0 \Rightarrow P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$

BASK 'in performansı daha iyi

Bant genişlikleri aynı

BASK'ta iki kaynak gerekiyor burda ise tek kaynak yeter.

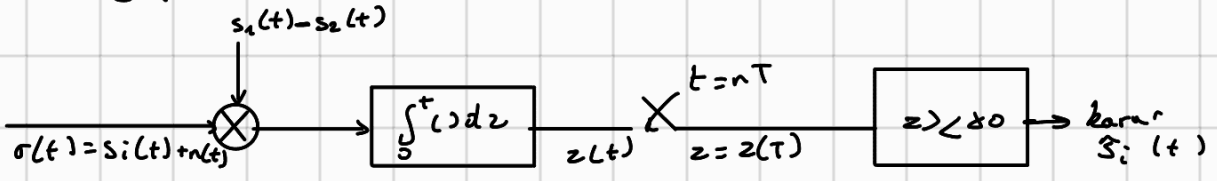
$\rightarrow E_b = \frac{A^2}{2}$

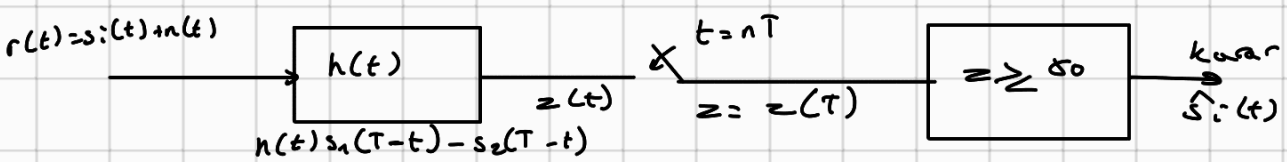


Zarf daha kötü (Sadece FSK'da sabit)

- İFKA, İFRKA, İGKA, VYA

Alıcı yapısı temel band haberleşmesi ile aynı,





İFKA (BPSK), İGKA (BASK)

$h(t) = s_1(T-t) - s_2(T-t) = 2s_1(T-t) = s_1(T-t)$

$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$

İFRKA (BFSK)

$h(t) = s_1(T-t) - s_2(T-t)$

$P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$

V-Y A (OOK)

$h(t) = s_1(T-t) - s_2(T-t) = s_1(T-t)$

$P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$

$P_b = Q\left(\frac{a_1-a_2}{2\sigma_0}\right) = Q\left(\sqrt{\frac{E_d}{2N_0}}\right) = Q\left(\sqrt{\frac{E_b}{N_0}(1-\rho)}\right)$ ✓

En mükemmel haberleşme beyin sinyalleri

Genlik - Faz kaydırmalı Anahtarlama

Amplitude - Phase Shift keying (ASK - PSK)

= Quadrature Amplitude Modulation (QAM)

Dörtlü Genlik Modülasyon (DÖGM)

$s_i(t) = \sqrt{\frac{2}{T}} A_i \cos(2\pi f_c t + \underbrace{\phi_i}_{a}), \quad 0 \le t \le T, \quad i = 1, \dots, M = 2^k$

$\phi_i = \frac{2\pi(i-1)}{M}$

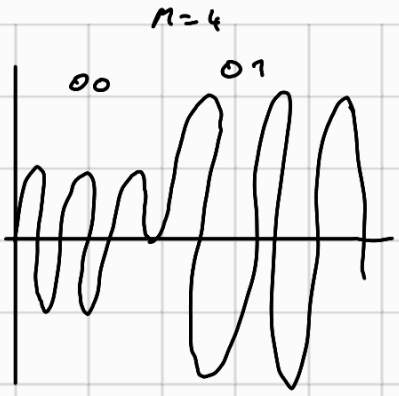
$= \sqrt{\frac{2}{T}} A_i \cos\phi_i \cos(2\pi f_c t) - \sqrt{\frac{2}{T}} A_i \sin\phi_i \sin(2\pi f_c t)$

ADSL Modemlerden önce kullanılıyordu

$M \uparrow \Rightarrow$ BG artar (BGV ↑) $\Rightarrow P_b \uparrow \Rightarrow$ BG verimli teknik

BGV Genlik k.A, PSK, QAM

Menzil kolmalı PSK, ve ASK dan daha iyi P_b ye sahiptir



Sınır 4kbit bakır kablo da

QAM modem hız 33kbit/sn

Trellis Coded Modulation (TCM) hem kaynak hem kanal kodama

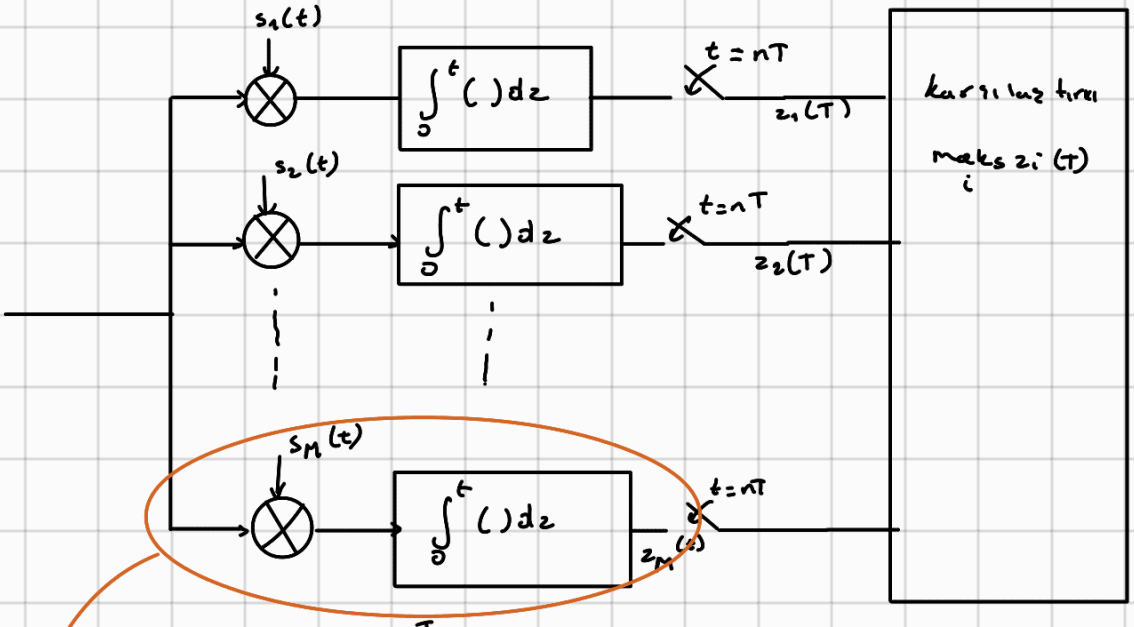
Modem hızı 56 kbit/sn

ADSL (OFTM'in muadili: Discrete Mulltitone Modulation)

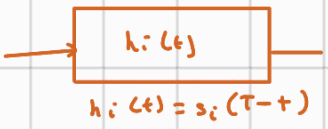
kablosuz kablolu

M'Li modülasyon teknikleri: isim

(AWGN kanalda) optimumalıcı



$$z_i(T) = \int_0^T r(t)\, s_i(t)\, dt \qquad i = 1, \ldots, M$$



M'li modülasyon işaretlerini vektör uzayında temsil etme

M psk

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \phi_i), \quad 0 \le t \le T, \quad i = 1, \dots, M = 2^k$$

$$= \sqrt{\frac{2E}{T}} \cos\phi_i \cos(2\pi f_c t) - \sqrt{\frac{2E}{T}} \sin\phi_i \sin(2\pi f_c t)$$

Tanım:

Temel fonksiyonlar (Basis function)

$$\psi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t), \quad 0 \le t \le T$$

$$\psi_2(t) = -\sqrt{\frac{2}{T}} \sin(2\pi f_c t), \quad 0 \le t \le T$$

$$\int_0^T \psi_1(t)\,\psi_2(t)\,dt = 0$$

$$E_{\psi_1} = E_{\psi_2} = \frac{2}{T} \cdot \frac{1}{2} \cdot T = 1$$

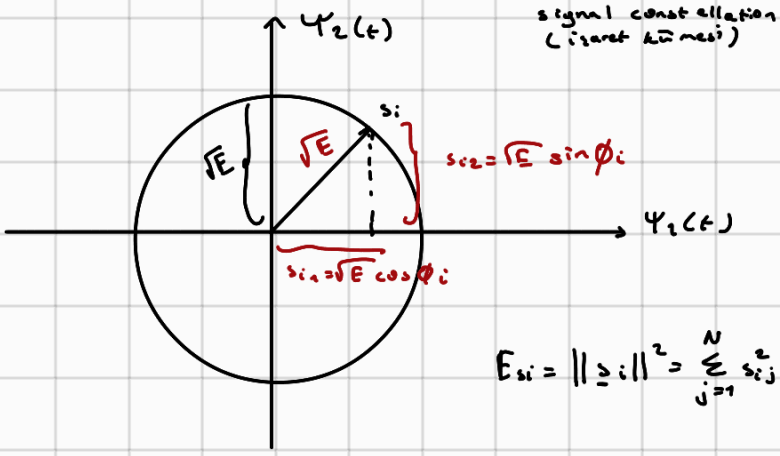
$\psi_1(t)$ ve $\psi_2(t)$ ortonormal işaretler

$$s_i(t) = \underbrace{\sqrt{E}\cos\phi_i}_{s_{i1}}\,\psi_1(t) + \underbrace{\sqrt{E}\sin\phi_i}_{s_{i2}}\,\psi_2(t) \quad \checkmark$$

M tane işaret ψ_1 ve ψ_2 dediğimiz ortonormal işaretlerin lineer kombinasyon şeklinde yazılabilir.

Vektörel olarak temsil edin

$$\underset{\text{vektör}}{\vec{s}_i} = \underbrace{[s_{i1}\ s_{i2}]}_{\text{uzayın eksenleri } \psi_1 \text{ ve } \psi_2 \text{ için}} = [\sqrt{E}\cos\phi_i \quad \sqrt{E}\sin\phi_i] \quad i = 1, \dots, M = 2^k$$

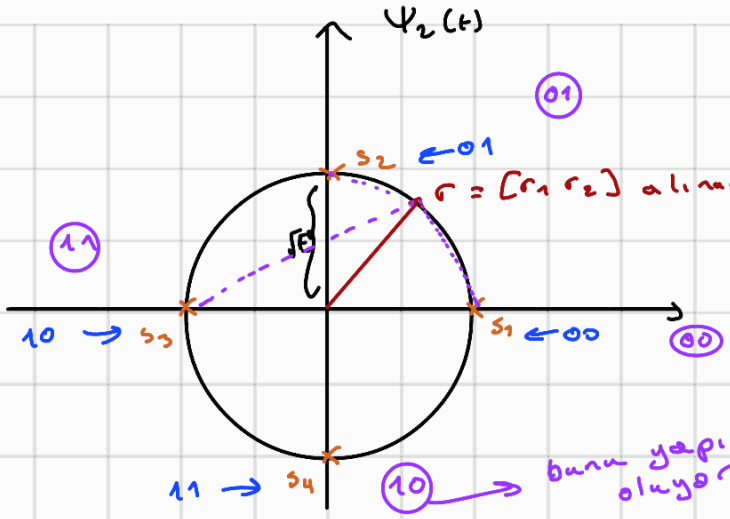


$$E_{si} = \|\vec{s}_i\|^2 = \sum_{j=1}^{N} s_{ij}^2$$

bu örnek için $E_{si} = \|s_i\|^2 = \sum_{j=1}^{N=2} s_{ij}^2 = s_{i1}^2 + s_{i2}^2 = E$ $\quad \|s_i\|$ = uzunluk

$M = 4 \quad (k = 2)$

i	$\phi_i = \frac{2\pi(i-1)}{M}$	
1	0	$\rightarrow s_1 = [\sqrt{E}\ \ 0]$
2	$\pi/2$	$\rightarrow s_2 = [0\ \ \sqrt{E}]$
3	π	$\rightarrow s_3 = [\sqrt{E}\ \ 0]$
4	$3\pi/2$	$\rightarrow s_4 = [0\ \ -\sqrt{E}]$



$r = [r_1\ r_2]$ alınan vektör

bunu yapınca bir sembolde bir bit hata oluyor

Gray kodlama

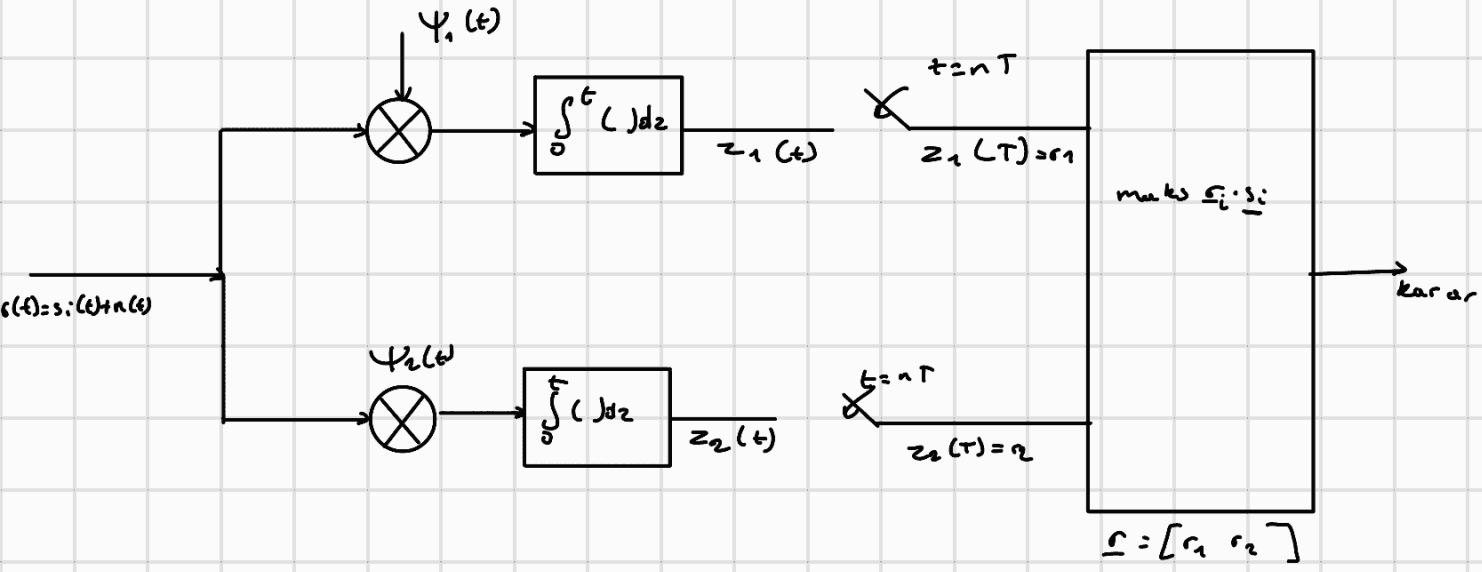
$P_b \cong \frac{1}{k} \cdot P_s$ → Sembol hata olasılığı

→ bit hata olasılığı

r hangisine daha yakınsa onlanmış olarak kararveriyor

bir sembollük hata 2 bitlik hataya karşılık geliyor.

s_2 gönderince en çok s_3 veya s_1'in yerine hata yapmaktır.



$x = [x_1\ x_2\ \cdots\ x_N]$

$y = [y_1\ y_2\ \cdots\ y_N]$

$x \cdot y = \sum_{i=1}^{N} x_i y_i$

iç çarpım (inner product)

FSK'ya uygulanamaz